Robogebra

SATURDAY · 10 OCTOBER · 04:00 PM IST · LIVE ONLINE

# *Still teaching maths* the hard way?

## The 10 Things That Tire Out Every Maths Teacher & How to Fix Them

Hosted by Muhammed Faiyaz, Robogebra
Expert panelist: Ashraf Sharfudeen

Keep a pen and paper ready. You'll need it in five minutes.

01 / 29 · Welcome
Chat: your name and city



THE PLAN

# *Your next* 60 minutes

01 **Find your score** · 20 min
10 moments that drain maths teachers. You rate each one.

02 **An easier way for every moment** · 15 min
Simple changes you can use in your very next period.

03 **The Vertex** · 5 min
Meet maths teachers from every board and state.

04 **Ask Ashraf, live** · 15 min
Bring the question from your own classroom.


Stay till the end: 2 weeks free

HOW TO SCORE

# *Write* 1 to 10 *down the side of your paper.*

For each moment, give it:

| *10* | *5* | *0* |
| --- | --- | --- |
| This is my week. I face it. | Sometimes, or not sure. | Not me. |

Add them up at the end. Your score stays with you.



Chat: type “ready”

1/10

# You search for good maths material online, and it looks **lifeless.**

Flat black-and-white pages. x^2 and sqrt(3x+1) typed as plain text. Steps that drift left and right, with equals signs that never line up.

YOUR SCORE | 10 | 5 | 0

```
Q. Solve: sqrt(3x+1) = x-1
 3x+1 = (x-1)^2
      3x + 1=x^2-2x+1
x^2-5x = 0
          x(x-5)=0
   x=0 or x = 5
 check: sqrt(16)=4 /true
```

**Would you show this to your class?**

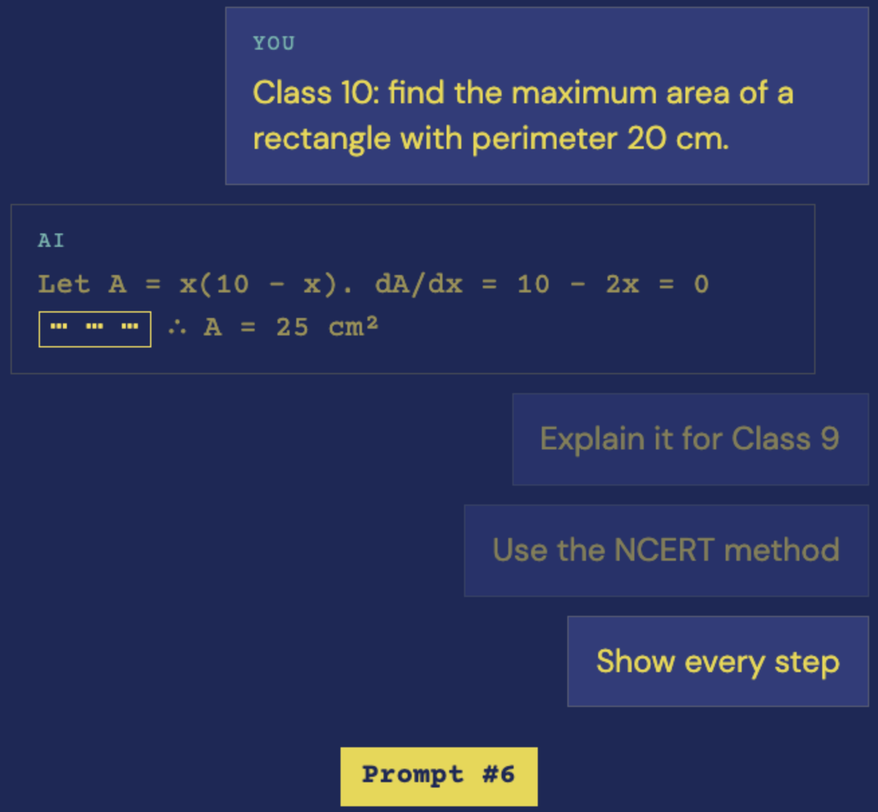

Write your score: 10 · 5 · 0

2/10

# *AI was supposed to save you time. Now you spend that time* fixing its answers.

Methods your syllabus never teaches. Steps skipped exactly where your students get stuck. Prompt after prompt, and it's 11 PM.

YOUR SCORE | 10 | 5 | 0 |





Write your score: 10 · 5 · 0

3/10

# *The tools are brilliant. Your period is* **40 minutes.**

Simulations, graphing apps and 3D models could make every concept click. But setup eats ten minutes, the bell rings mid-demo, and the syllabus is already two chapters behind.

YOUR SCORE | *10* | *5* | *0*




- 0–5 min · settle, attendance
- 5–15 min · find the tool, set it up
- 15–40 min · demo… the bell rings mid-demo

Syllabus: 2 chapters behind · Unit test: next week

**Still loading…**



Write your score: 10 · 5 · 0

4/10

# *You sit down to prepare one lesson. Two hours later, you're still* **learning the tool.**

Toolbars full of icons. Menus inside menus. A 12-part tutorial series. Your lesson plan is still a blank page.

YOUR SCORE | 10 | 5 | 0 |



Write your score: 10 · 5 · 0

▶ Getting started: Part 1 of 12 · 18:42

▶ Part 2: The toolbar explained · 22:10

▶ Part 3: Settings and preferences · 15:35

*Lesson plan: Circles*

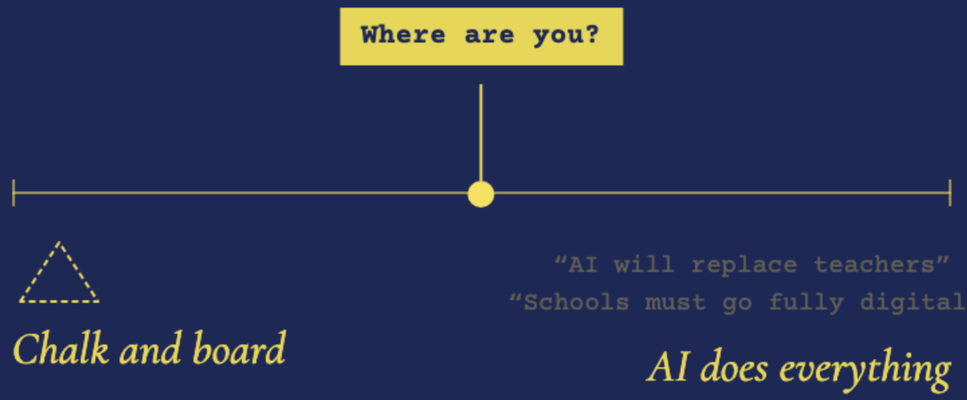
Still blank

# 5/10

## *Where do you stand on teaching with technology? Most days,* **you're not sure.**

One side: AI headlines promising to change everything, and warning that teachers will be replaced. Other side: chalk and board, which you trust, struggling to hold a class that grew up on reels.

YOUR SCORE | 10 | 5 | 0 |





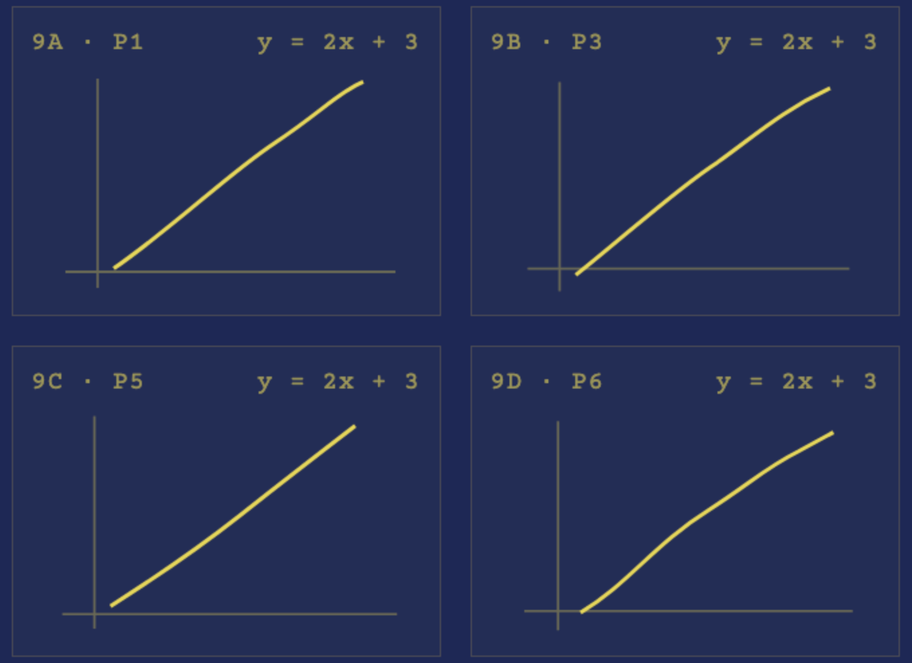
Write your score: 10 · 5 · 0

6/10

*Your classroom has a smart board. You still write the same sum on it* **five times a day.**

Period 1 in 9A: you draw the graph. Period 3 in 9B: the same graph, from scratch. Period 6 in 9D: all over again, and your hand aches by the end of the day.

YOUR SCORE | 10 | 5 | 0 |




4th time today


Write your score: 10 · 5 · 0

7/10

# *One good question. Your students need five more like it.* Out come the guidebooks.

Three reference books, two guides and a pile of old question papers. An hour later you have four variations, and one of them has an answer that won't come out clean.

YOUR SCORE | 10 | 5 | 0 |

10 / 29 · Find your score
Write your score: 10 · 5 · 0

## *Q1 variations*

1. $x^2 + 5x + 6 = 0$ ✓

~~2. $2x^2 + 3x - 9 = 0$ (wrong chapter)~~

~~3. $x^2 - 4x + 1 = 0$ (too easy)~~

4. $3x^2 + 7x - 4 = 0$ ✓

5. $x = 17 / 13\sqrt{2}$ ??

3 reference books · 2 guides · old papers · 1 hour

Just one more…

8/10

# *Thirty-nine heads nod. One doesn't. That student still* **can't picture the cone.**

A frustum in Class 10. A line cutting a plane in Class 12.
On a flat board, a 3D idea shrinks to a few slanted lines,
and some students never see what you see in your head.

YOUR SCORE | 10 | 5 | 0 |


Write your score: 10 · 5 · 0



9/10

# *Exams are next week. The question paper is* due on Friday.

The right 3-marker is in one guide, the case study in an old question paper, the diagram in a book a colleague borrowed. Then it all has to match the blueprint, fit the marks, and get typed up.

YOUR SCORE [10] [5] [0]

12 / 29 · Find your score
Write your score: 10 · 5 · 0

| CHAPTER | 1M | 2M | 3M | 5M | CASE |
|---|---|---|---|---|---|
| Real Numbers | ✓ | ✓ | ✓ | – | – |
| Triangles | ✓ | ? | ✓ | ✓ | – |
| Circles | ✓ | ✓ | ? | – | ? |
| Statistics | ✓ | ✓ | ✓ | ✓ | – |
| Probability | ? | ✓ | – | – | ✓ |

Q14?

need 1 more case study

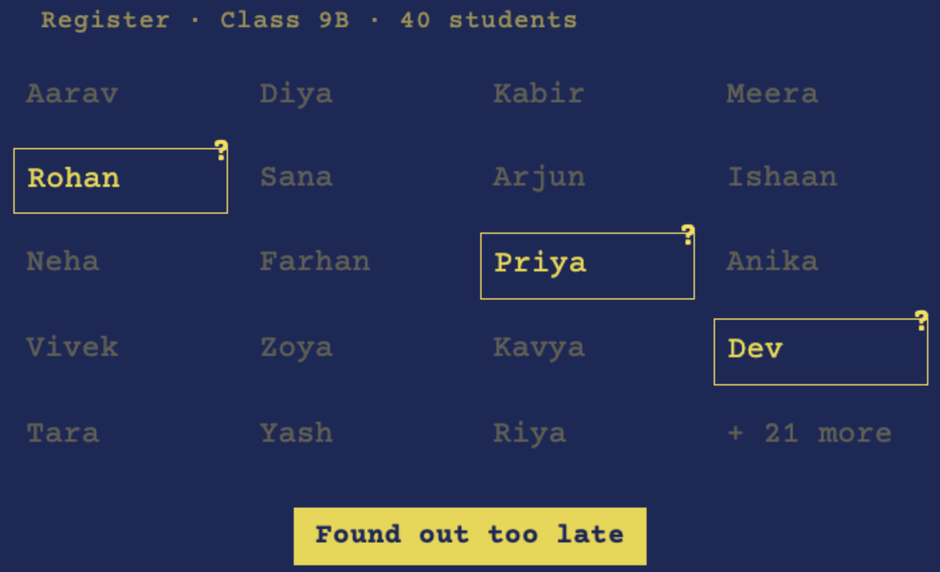
Due Friday

# *10*/10

## *Forty students. Forty minutes. You only find out who's lost* **when the test papers come back.**

Rohan keeps losing minus signs. Priya freezes on word problems. Someone at the back hasn't understood fractions since Class 6. Finding out who needs what takes time your period doesn't have.

YOUR SCORE | *10* | *5* | *0*






Write your score: 10 · 5 · 0

YOUR SCORE

# *Add up your* 10 numbers.

| | |
|---|---|
| *0 to 30* | You've found your rhythm. The easier ways ahead are handy extras to keep in your pocket. |
| *35 to 65* | Some of these moments are eating into your week. A few of the easier ways ahead will give you real time back, every week. |
| *70 to 100* | You carry most of these moments, week after week. This session was built for you, so write down every easier way you see. |

Your score stays with you.


Sharing in chat is optional

WHO WE ARE

# *For 15 years, we've chased one simple question: how can maths be* **easier to teach, and to learn?**

robocompass

## *Robocompass*

Our free maths tool, used by lakhs of students and recommended by NCERT. Ranked the **3rd most-used ICT tool** in Indian schools (Kotak Mahindra Bank impact assessment study).

UNIVERSITEIT GENT

## *Ghent University, Belgium*

We built maths teaching simulations together.

PhET INTERACTIVE SIMULATIONS University of Colorado Boulder

## *PhET Interactive Simulations*

University of Colorado Boulder, founded by Nobel laureate Carl Wieman. We worked on PhET sims.



Chat: used Robocompass? yes / no



Expert panelist

MEET OUR EXPERT PANELIST

# *Ashraf* Sharfudeen

- ✓ Founder of Robogebra and Provility
- ✓ Creator of Robocompass and MathDisk
- ✓ Co-author of a peer-reviewed paper on progressive disclosure in maths teaching, with Dr. Ajit Kumar and Dr. Sang-Gu Lee

Ask him anything at the end of the session.


Ashraf says hello

EASIER WAY 1

for “Maths online looks lifeless”

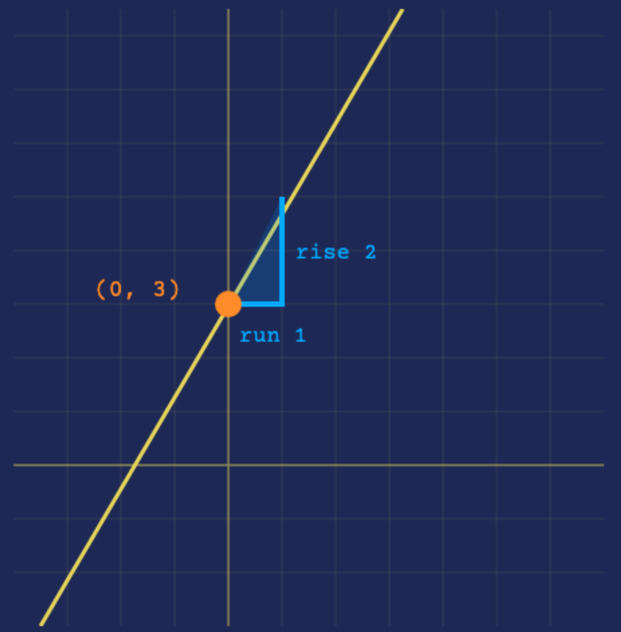
# *Give each idea its own colour, and keep that colour* **everywhere it appears.**

In y = mx + c, if m is blue in the equation, the slope is blue on the graph. If c is orange, the point where the line meets the y-axis is orange too. Students follow the colour and see where each part comes from.



before: y=2x+3 , m=slope , c=y-intercept

$$y = 2x + 3$$




**Same maths, now in colour**

EASIER WAY 2

for “AI answers that miss your syllabus”

# *AI only knows what you tell it. Give it your class, your textbook and your method* in the first prompt.

Coming soon in The Vertex: a free prompt library for maths teachers.



Solve $x^2 - 5x + 6 = 0$ → $x = (5 \pm \sqrt{1}) / 2$ …

A PROMPT THAT WORKS

> I teach Class 10, CBSE, NCERT Maths, Chapter 4: Quadratic Equations. Solve this question by factorisation only, the method in the textbook. Don't use the quadratic formula. Show every step on a new line, with one simple sentence explaining it, for a Class 10 student.

Class | Board | Textbook and chapter | Method allowed | How many steps | Who it's for

First try

EASIER WAY 3

for “Brilliant tools, 40-minute periods”

# *Keep the tool's setup* out of your period.

01 **Pick tools that follow your syllabus.** When a tool is organised by your chapters, the right screen is one tap away.

02 **Prepare it the night before.** If your tool isn't organised that way, set up the exact screen you need, save it, and keep it ready to open on the smart board.

03 **Dry-run it three times.** Walk through the lesson on the tool at least three times before you use it with a class.



BEFORE



- [ ] 10 min · open laptop, find tool, adjust sliders
- [ ] the bell rings mid-demo

AFTER



- [x] 30 sec · tap saved lesson
- [ ] the rest · teaching and student questions

✓ Follows my chapter ✓ Saved and ready ✓ Dry-run ×3

**Ready in 30 seconds**

EASIER WAY 4

for “Learning the tool before the lesson”

# *Before you try any tool, ask one question:* **how much will I have to learn** *before I can teach with it?*

Robogebra sits at the low end. You click a button, and that's all you need to know to use it in class.

THE LEARNING COST SCALE

| | |
|---|---|
| *High* | Tutorials, toolbars and settings before your first lesson |
| *Medium* | A few hours of practice, then it gets easier |
| *Low* | Open it and teach |

**One button**

EASIER WAY 5

for “Stuck between chalk and AI”

# *AI will keep changing. A teacher explaining an idea to a class* **won't.**

Stand in the middle: keep your syllabus, your textbook and your way of teaching, and add tools that make them more visual. Before you adopt any tool, check three things:

- ✓ Nothing new to learn
- ✓ No change to how your school works
- ✓ Works with your board and textbook

Robogebra works on the smart board you already have and fits the way you already teach.




*Chalk and board*

“AI will replace teachers”
“Schools must go fully digital”

*AI does everything*

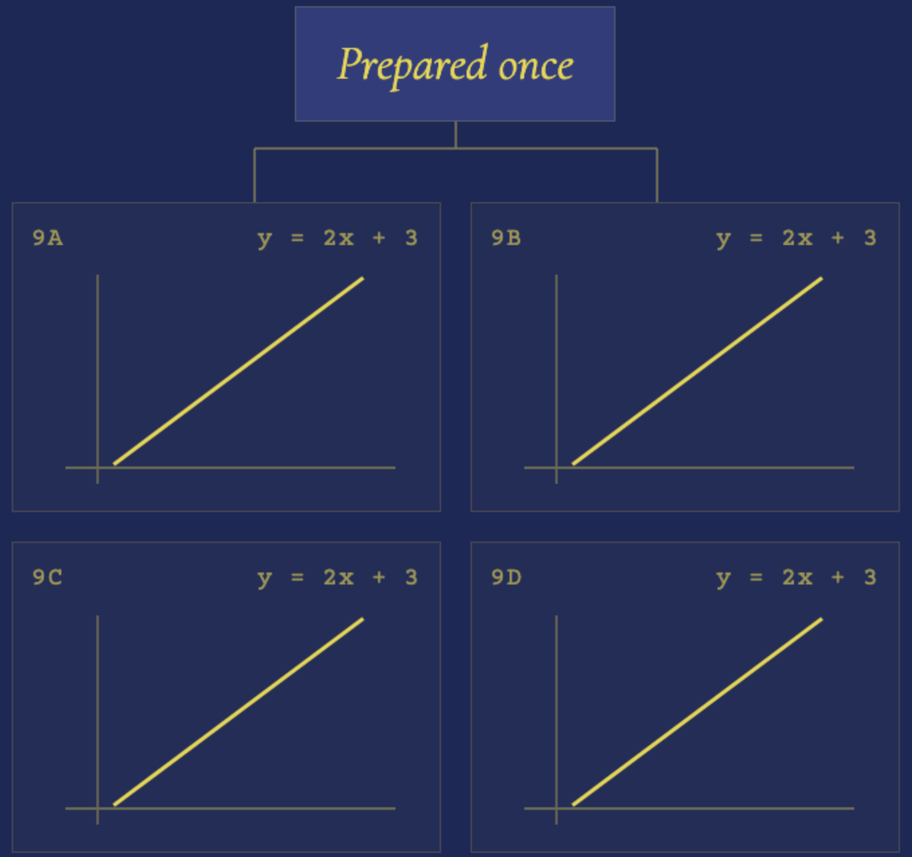
EASIER WAY 6

for “A smart board, used like a blackboard”

# *Prepare the lesson once, and* open it in every section.

01 **Do it yourself with AI:** turn your lesson into a simple web page and open it on the smart board in 9A, 9B, 9C and 9D. Coming soon in The Vertex: a step-by-step playbook for doing this with AI.

02 **Or let Robogebra do it:** pick an example or exercise, start the lesson, and the solution appears step by step, as if you were writing it on the board.




**Drawn once. Used four times.**

EASIER WAY 7

for “Hunting for one more practice question”

# *One good question can give you ten more.* **Change the numbers** *and see the new solution straight away.*

01 In Robogebra's **Edit and Explore**, you change an expression or a coefficient, and the built-in maths engine solves the new version live, step by step.

02 Other tools can do this too. Look for one that lets you edit the expression and solves it for you.

03 Use it live in class, or to prepare practice sets and homework in minutes.



~~$x^2 - 5x + 6 = 0$~~

$$x^2 - 7x + 12 = 0$$

$$(x - 3)(x - 4) = 0$$

$$x = 3 \text{ or } x = 4$$

**New question, solved live**

EASIER WAY 8

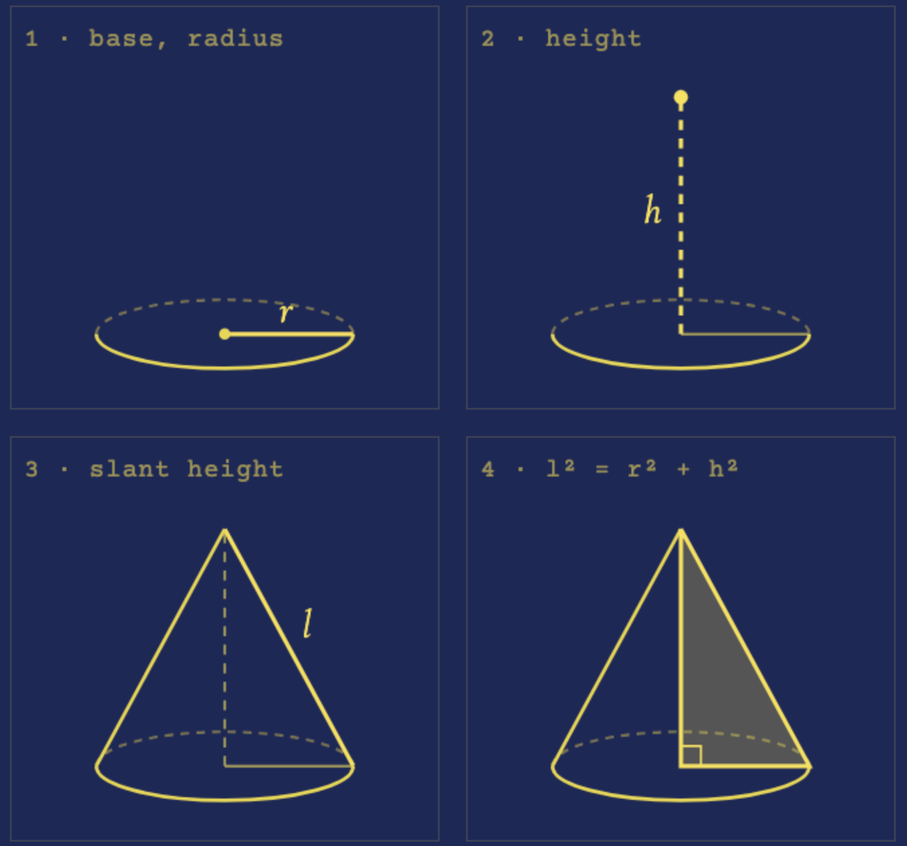
for “The student who can't picture the cone”

# *Let students watch the shape being built,* **one piece at a time.**

01 **With AI:** you can now create 3D visuals from a simple prompt. The Vertex prompt library will include prompts for these too.

02 **With Robogebra:** for any textbook question that needs a 3D or 2D picture, the model builds up step by step, alongside the solution.

One piece at a time, the student builds a clean picture in their head.




**Now I see it**

EASIER WAY 9

for “Setting the question paper”

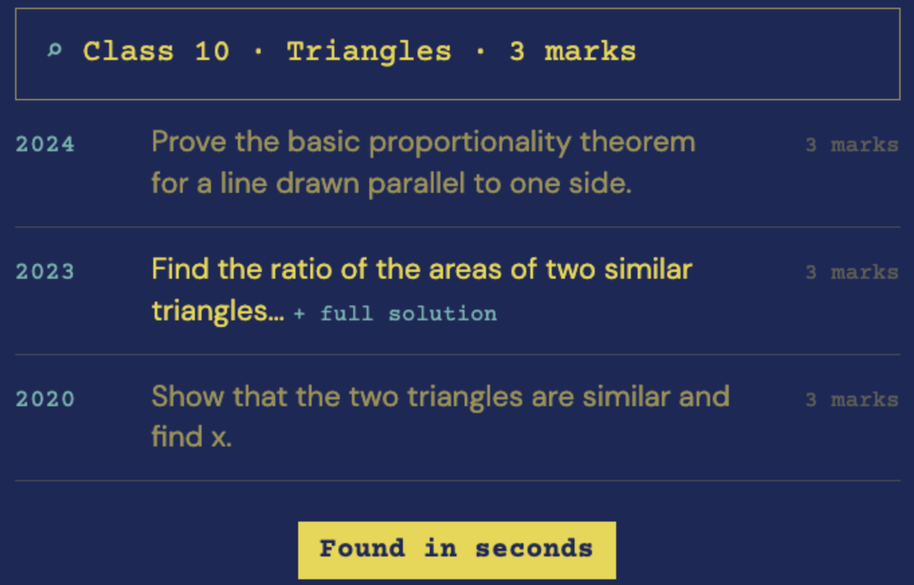
# *Keep every good question in* one place you can search.

01 **Build your own with AI:** photograph or scan your reference books and old question papers, and use AI to turn them into a searchable collection, tagged by chapter, marks and question type.

02 **Or start with Robogebra:** previous years' questions are already there, and you can search them.

Search “Class 10 · Triangles · 3 marks” and find what you need in minutes.





EASIER WAY 10

for “Who needs help, and with what?”

# *Let students answer on the same platform* where you set the questions.

Dictating questions in class, or sending them on WhatsApp, leaves the answers scattered in forty notebooks and chats. When students answer on one platform, you can see who is stuck, where, and for how long.

01 **In Robogebra:** you unlock a question, and it reaches your students. Then you see each student's journey through the textbook: where they got stuck, and where they're taking longer.

02 Use a platform that works this way, or build one.



Unlock → question reaches students → their journey

| | |
|---|---|
| *Rohan* | method right · **lost the sign**, step 3 |
| *Priya* | **word problems** · 6 min before starting |
| *Dev* | stuck on the **fractions** step, again |
| *37 more* | on track |

**Found out in time**

THE VERTEX

# *The point where many lines meet,* where maths teachers from every board and state meet.

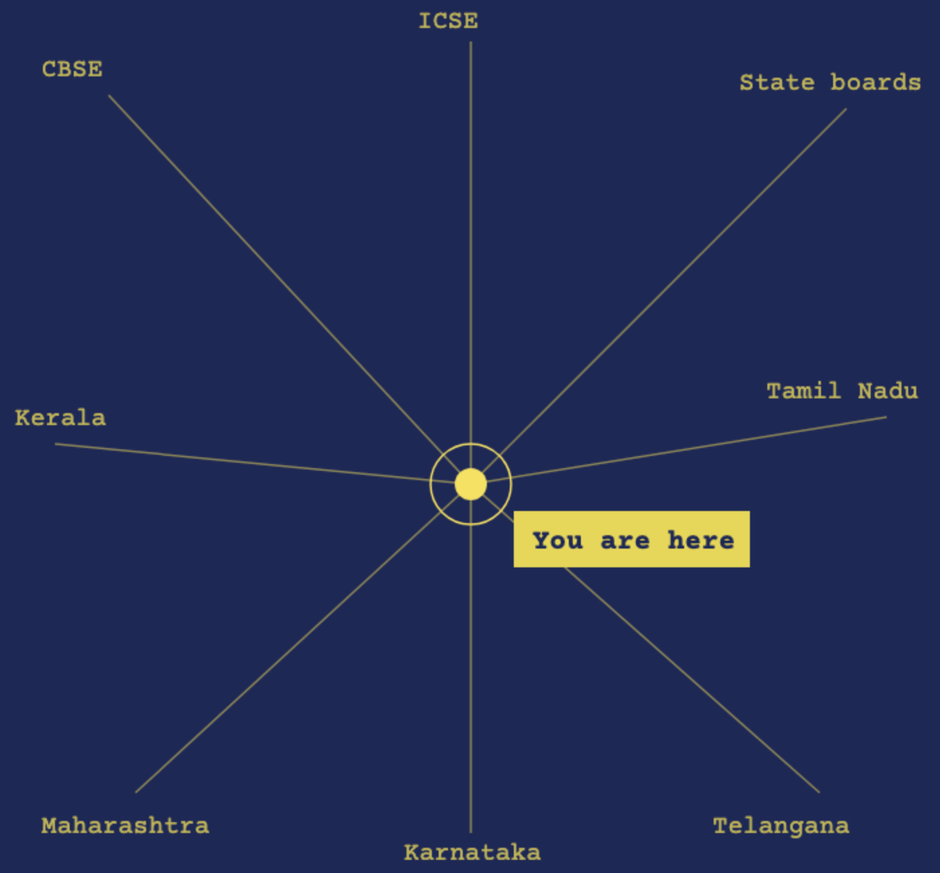
What's waiting for you inside:

01 The maths teacher prompt library, for lessons, solutions and 3D visuals

02 The step-by-step playbook for turning your lessons into web pages with AI

03 Teachers from across India sharing what works in their classrooms, and what doesn't

Everything we've promised today arrives in The Vertex this week.





Post your intro in the group



Expert panelist

ASK ASHRAF, LIVE

# *Your classroom,* your question.

Type your question in the chat, and start it with **“Q:”**

Not sure what to ask? Try:

01 Which of today's moments do you want help with first?

02 How would one of these easier ways work for your class, board or chapter?

03 Something in your classroom that we didn't cover today



Start your question with “Q:”

Robogebra

# *Thank you for your* Saturday evening.

You came live, so your 2 weeks of free Robogebra access is yours.

We'd love to hear from you. Share one thing you'll try in your very next period.

✓ Nothing new to learn · ✓ No change to how your school works · ✓ Works with every board and textbook



Chat: one thing you'll try next period